

Marwan Mohamed Elfouly

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github.com/Marwan-Mohamed14 | marwan-mohamed14.github.io/portfolio

EDUCATION

Misr International University

Cairo, Egypt

Bachelor of Science (B.Sc.), Computer Science — Software Engineering Track

Expected: May 2026

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Software Engineering, Database Systems, Machine Learning, Object-Oriented Programming, Computer Networks, Software Design Patterns, ISTQB Foundation Level

EXPERIENCE

Cybersecurity Intern

July 2025 – September 2025

Siemens

Cairo, Egypt

- Evaluated OT/ICS security architectures across 3+ scenario exercises, identifying vulnerabilities using Siemens industrial security frameworks
- Conducted pen testing on live PLC hardware using Kali Linux and solved 30+ ICS-focused CTF challenges in an industrial security environment

Instructor

December 2024 – June 2025

Waleed Academy

Cairo, Egypt

- Designed CS curriculum and produced 75+ bilingual (Arabic/English) video lessons on algorithms, OOP, and Linux, published across multiple online learning platforms
- Published Linux administration (beginner to advanced) and Selenium automation playlists reaching learners across Arabic-speaking markets

Odoo Developer Intern

August 2024 – September 2024

Siemens

Cairo, Egypt

- Developed and customized Odoo ERP front-end modules in Python and JavaScript, streamlining data entry workflows for the operations team
- Optimized MySQL queries across internal reporting pipelines, reducing data retrieval overhead for operational dashboards

SQL Developer Intern

July 2024 – August 2024

Summit Technology Solutions

Cairo, Egypt

- Designed normalized MySQL schemas supporting client reporting workflows; resolved slow-query bottlenecks through indexing and query restructuring

PROJECTS

Graduation Project – AI Contract Validation System | *React, Spring Boot, OCL, MySQL* |

github.com/Marwan-Mohamed14/softwareguard-graduation-project

- Architected Spring Boot microservices + React frontend for contract validation, decomposing into document ingestion, OCL rule evaluation, and JWT-based authentication modules
- Developed RESTful APIs with role-based authorization and integrated an OCL rule engine with an AI model to automatically detect legal issues in contract clauses

AI Fitness Trainer | *Flutter, Dart, Groq AI (LLaMA 3.3 70B), Supabase* |

github.com/Marwan-Mohamed14/AI_Fitness_Trainer

- Built a cross-platform mobile app generating personalized weekly workout and diet plans using Groq AI based on user goals, age, preferences, and dietary restrictions
- Implemented daily meal/workout logging, a visual monthly progress dashboard, gym discovery via Google Places API, and a real-time social community feed powered by Supabase

Football Analytics Dashboard & Match Prediction Model | *Power BI, Python, Scikit-learn* |

github.com/Marwan-Mohamed14/Ai_model_matches_prediction

- Built Power BI dashboard analyzing 58,000+ match records across global clubs, enabling data-driven scouting through 15+ tactical profiling metrics including pace, passing, and defensive output
- Trained Random Forest and Decision Tree models to predict match outcomes and expected goals from rolling 5-match windows and betting odds from 8+ bookmakers

TECHNICAL SKILLS

Programming Languages: C++, Python, Java, JavaScript, Dart, C#, HTML/CSS

Frameworks: Spring Boot, Flutter, Node.js, React, Odoo

Databases: MySQL, Firebase, Supabase, MongoDB

Developer Tools: Git, Power BI, Selenium

AI / ML: Fine Tuning, Data Preprocessing, Machine Learning, Agents, OCL Rule Engine

Systems: Linux, Kali Linux

CERTIFICATIONS

Flutter Developer Certificate — Digital Egypt Pioneers Initiative (DEPI), MCIT	2025
Software Security Course — IBM	2025
Siemens Transformena Attendance Siemens	2025

LANGUAGES

Arabic (Native) | English (C1) | German (B1)